

How to Prepare for an R Quiz

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March 31, 2026

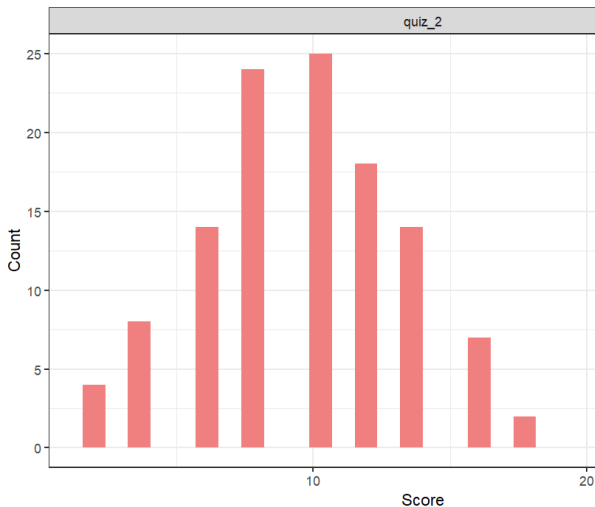
Outline

- 1 About the Grades
- 2 7 Sources of R Quiz Problems
- 3 How to Prepare for an R Quiz

Quiz 2 grade distribution

- Mean = 9.9, SD = 3.9, Q1 = 8, Median = 10, Q3 = 12

Quiz Score Distributions



Will low score affect your final grade?

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Will low score affect your final grade?

- "Our grading is curved. Your letter grade will be based on your percentile rank in class".

Will low score affect your final grade?

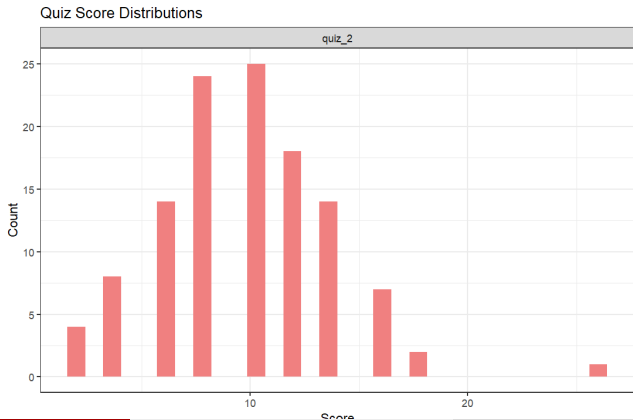
- "Our grading is curved. Your letter grade will be based on your percentile rank in class".
 - Percentile is more important than the score.

Will low score affect your final grade?

- "Our grading is curved. Your letter grade will be based on your percentile rank in class".
 - Percentile is more important than the score.
- One exception . . . if you are the outlier.

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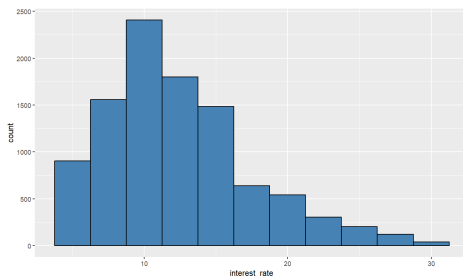


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Source 1: Past Exams

- Problems belong to this source: **Q1, Q4, Q5**.
- **Q4.** Which of the following code snippets would most likely produce the graph below?



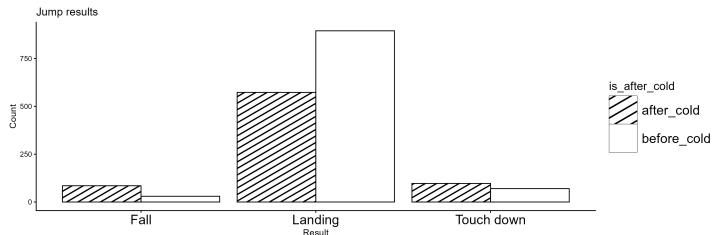
A.

```
ggplot(loan, aes(x = interest_rate)) +  
  geom_histogram(binwidth = 2.5, fill = "steelblue")
```

Source 2: Slide Contents

- Problems belong to this source: **Q3, Q12**.
- Q12.** What will `medalist_bar_plot` look like if we remove Lines 9 and 11, as well as the `%>%` in Line 10?

D.



Source 3: Course

- Some additional information may not be placed in the slides.
 - Otherwise the slide will be too crowded.
- Problems belong to this source: **Q2, Q7**.
- **Q2.** Which of the following statements is FALSE?
 - A. `setwd("C:\\your\\working\\directory")` does not work on Mac.
 - B. We can use `df[["col"]] <- c(1, 2)` to add a new column to `df`.
 - C. `t.test(df$weight ~ df$habit)` performs an independent two-sample t-test.
 - D. The main reason to use `install.packages()` only when needed is to prevent function name conflicts.

Source 4: Statistics/Econometrics

- Problems belong to this source: **Q9, Q10**.
- Q10**. Consider the following three ways to calculate the p-value:

```
test_1 <- t.test(data$not_Yuriko_appear_rate, mu = 1)
p_1 <- test_1$p.value
test_2 <- t.test(data$not_Yuriko_appear_rate, data$Yuriko,
                 paired = TRUE)
p_2 <- test_2$p.value
test_3 <- t.test(data$not_Yuriko_appear_rate, data$Yuriko)
p_3 <- test_3$p.value
```

Which of the following statements best describes the relationship among p_1 , p_2 , and p_3 ?

A.

$$p_1 = p_2 = p_3$$

Source 4: Statistics/Econometrics

- $x_i = \text{not_Yuriko_appear_rate}_i$, $y_i = \text{Yuriko}_i = 1$, ($i = 1, \dots, n$).

-

$$t_1 = \frac{\bar{x} - 1}{SE_{\bar{x}}}$$

$$t_2 = \frac{\bar{x}_{\text{diff}} - 0}{SE_{\bar{x}_{\text{diff}}}} = \frac{\bar{x} - 1}{SE_{\bar{x}-1}} = \frac{\bar{x} - 1}{SE_{\bar{x}}} = t_1$$

$$t_3 = \frac{\bar{x} - \bar{y}}{SE_{\bar{x}-\bar{y}}} = \frac{\bar{x} - 1}{SE_{\bar{x}-1}} = \frac{\bar{x} - 1}{SE_{\bar{x}}} = t_1$$

- Hence $t_1 = t_2 = t_3$, and $p_1 = p_2 = p_3$.

Source 5: Practice Mistakes

- Problems belong to this source: **Q6, Q8**.
- **Q8.** Which of the following statements is FALSE?
 - A. The code still works if we replace `library(tidyverse)` with `library(dplyr)` in Line 1.
 - B. The code still works if we replace `gnosia_tidy[1:i, ]` with `gnosia_tidy[1:i]` in Line 22.
 - C. The code still works if we replace `pval < 0.0001 && finish_index == 0` with `pval < 0.0001 & finish_index == 0` in Line 25.
 - D. Creating a new function `t_test_pval` is not necessary for the code to work. That is, there is an equivalent way to write the code without using `function()` and still obtain the same result as the original code.

Source 5: Practice Mistakes

• **Q6.** Which of the following statements is FALSE?

A. The following code works for (Code 1).

```
data$appear_sum <- data %>% rowSums()
```

B. The following code works for (Code 1).

```
data <- data %>% mutate(appear_sum = rowSums())
```

C. The following code works for (Code 2).

```
data$not_Yuriko_appear_rate <- (data$appear_sum -  
                                data$Yuriko) / 12
```

D. The following code works for (Code 2).

```
data <- data %>% mutate(not_Yuriko_appear_rate =  
                        (appear_sum - Yuriko) / 12)
```

Source 6: Things Learned when Doing Practice

- Problems belong to this source: **Q13, Q15.**
- **Q15.** Which of the following statements about `ggsave()` is correct?
 - A. We can save two plots at the same time if they have the same path, width, height, and dpi settings.
 - B. In Lines 51 to 56, changing `width = 12, height = 8` to `width = 6, height = 4` not only makes the saved image smaller, but also changes the layout of the graph.
 - C. If the folder specified in `path = "output"` does not exist, `ggsave()` immediately returns an error and says it cannot find the directory.
 - D. If we do not specify the file type to save (for example, `filename = "medalist_hist_plot"`), it will be saved as a JPEG file by default.

Source 6: Things Learned when Doing Practice

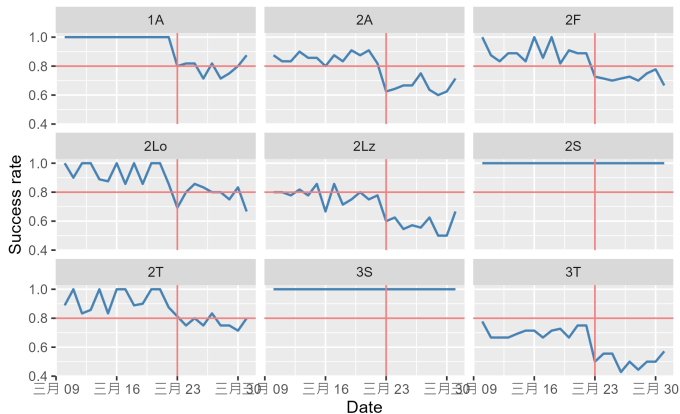
- Plot when `width = 12`, `height = 8`:



Source 6: Things Learned when Doing Practice

- Plot when `width = 6`, `height = 4`:

Daily landing success rate (by jump)



Source 7: Answer Code(s)

- There are a lot of ways to do the practices.
- Problems belong to this source: **Q11, Q14**.
- **Q14**. Which of the following statements is FALSE?

A. The code below is an equivalent way to generate `medalist_line_plot`:

```
medalist_line_plot <- medalist_tidy %>%
  group_by(date) ...
```

- B. Placing `%>%` at the end of each line not only improves readability, but also prevents errors that would occur if `%>%` were moved to the beginning of the next line.
- C. Forgetting to call `ungroup()` in Line 32 will cause a problem here: it affects the output of `medalist_facet_line_plot`.
- D. We can merge all `mutate()` to a single one from Line 7 to 11:

```
medalist_tidy <- medalist_raw %>%
  mutate( ...
```

Question Sources Table

- Memorizing only slide contents and past exams is not enough ...

Source	Questions
Source 1: Past Exams	Q1, Q4, Q5
Source 2: Slide Contents	Q3, Q12
Source 3: Course	Q2, Q7
Source 4: Statistics/Econometrics	Q9, Q10
Source 5: Practice Mistakes	Q6, Q8
Source 6: Things Learned when Doing Practice	Q13, Q15
Source 7: Answer Code(s)	Q11, Q14

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- Practice makes perfect.

Things to mentioned

- 2 more R Quiz in this semester: Quiz 4 (May 5), Quiz 6 (May 26).